Uni. Roll No:-.....

SET-C

GLA University, Mathura

Course: -B.Tech. II-Year, End-Term (Even Sem.) Examination, 2012-13

Subject: Database Management System (ITE 221)

Time:-2.30 Hour M.M-40

Notes:-

1.Answer all questions from Group A (which is) compulsory, Any Four from Group B and Any Four from Group C.

2.All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.

3.Answer should be brief and to-the-point and be supplemented with neat sketches.

4.Any missing or wrong data may be assumed suitably giving proper justification.

5.Figures on the right-hand side margin indicate full marks.

Group-A (All questions are compulsory) 16X0.5=08

1. The DBMS acts as an interface between what two components of an enterprise-class database system?
   - Database application and the database
   - Data and the database
   - Database application and SQL
   - The user and the database application

2. The following are components of a database except _____.
   - user data
   - reports
   - metadata
   - Indexes

3. Every time attribute A appears, it is matched with the same value of attribute B, but not the same value of attribute C. Therefore, it is true that:
   - $A \rightarrow B$.
   - $A \rightarrow (B,C)$.
   - $(B,C) \rightarrow A$.
   - $A \rightarrow C$.

4.A relation is in this form if it is in BCNF and has no multivalued dependencies:
   - second normal form.
   - third normal form.
   - domain/key normal form.
   - fourth normal form.

5.Row is synonymous with the term:
   - record.
   - .column.
   - relation
   - field.

6. Which type of entity cannot exist in the database unless another type of entity also exists in the database, but does not require that the identifier of that other entity be included as part of its own identifier?
   - Weak entity
   - Strong entity
   - ID- independent entity
   - ID-dependent entity

7. Every view serializableschedule which is not conflict serializable has
   - Dirty writes
   - Conflicting writes
   - blind writes
   - uncommitted writes

8. It is possible to define a schema completely using
- VDL and DDL.
- DDL and DML.
- VDL and DML.
- SDL and DDL

9.To delete a particular column in a relation the command used is:
- UPDATE
- ALTER
- DROP
- DELETE

10. The _____ operator is used to compare a value to a list of literals values that have been specified.
- BETWEEN
- ANY
- ALL
- IN

11. Which of the following is the preferred way to recover a database after a transaction in progress terminates abnormally?
- Rollback
- Rollforward
- Reprocess transactions
- Switch to duplicate database

12. Concurrency control is important for which of the following reasons?
- To ensure data integrity when updates occur to the database in a multiuser environment
- To ensure data integrity while reading data occurs to the database in a multiuser environment
- To ensure data integrity when updates occur to the database in a single-user environment
- To ensure data integrity while reading data occurs to the database in a single-user environment

13. Locking may cause which of the following problems?
- Erroneous updates
- Versioning
- Deadlock
- All of the above.

14. Performance analysis and tuning is which of the following?
- Undertaken one time at the time of a DBMS installation
- Undertaken as an ongoing part of the backup of a database
- Undertaken one time at the time of the implementation of a new application
- Undertaken as an ongoing part of managing a database

15. Backward recovery is which of the following?
- Where the before-images are applied to the database
- Where the after-images and before-images are applied to the database
- Where the after-images are applied to the database
- Switching to an existing copy of the database

16. Which of the following is true concerning a global transaction?
- The required data are at one local site and the distributed DBMS routes requests as necessary..
- The required data are at one local site and the distributed DBMS passes the request to only the local DBMS
- The required data are located in at least one nonlocal site and the distributed DBMS routes requests as necessary
- The required data are located in at least one nonlocal site and the distributed DBMS passes the request to only the local DBMS.

Group-B (Answer any Four)4x3=12

Q1 What is normalization? Write its advantages and disadvantages. (1+1+1)

Q2 Draw the diagram for the overall Database Structure?

Q3 Under what conditions views are updateable?

Q4 Write a command for creating table with following constraints –

BankInfo Table

Colum name	Datatype	Description	Constraints	
AccountNo	Number(10)	Account no of customer		
CustomerId	Varchar2(6)	Unique id provided to each customer when he/she is registered to purchase items	Foreign key referring to customer table	Composite Primary key

Q5 Write short notes on(any one) –
(a) Transaction properties (b) Explicit Cursor

Group-C (Answer any Four) 4x5=20

Q1 Explain Deadlocks. Discuss the deadlock prevention schemes used in databases. (2+3)

Q2Which is the most important structure that is used to recover from database failures? Explain the update techniques to maintain log files in detail. (1+4)

Q3What are distributed databases? How failures are handled in the distributed environment? (2+3)

Q4What are the goals for concurrency control? Discuss the popular protocol used for concurrency control? (2+3)

Q5 Explain the concept of serializability? Also, draw the dependency graph of the following schedule and check whether it is conflict serializable? (2+2+1)

T1 :	R(A),W(A)		R(B),W(B)	
T2 :		R(A),W(A)		R(B),W(B)

GLA University, Mathura **SET-A**

END Term Examination, 2012-13

Course: - B.Tech (CS) **II-Year, IV-Sem.**

Subject: - Introduction to Microprocessor (CSE 222) **Uni. Roll No:-**

Time:-2 hours 30 Minutes **Total Marks:-40**

Notes:-

1) *Answer all questions; Section A is **compulsory**, **Any four** from Section B and **Any four** from Section C.*
2) *All parts of a question (a, b, etc.) should be answered at one place.*
3) *Answer should be brief and to-the-point and be supplemented with neat sketches.*
4) *Any missing or wrong data may be assumed suitably giving proper justification.*
5) *Figures on the right-hand side margin indicate full marks.*

SECTION-A

Objective Questions (Write the correct answer): **(16*0.5 = 8)**

I. The act of acquiring an instruction is referred as the____ instruction.

- Fetching
- Fetch cycle
- Decode
- Execution

II. The subprogram finish the return instruction recovers the return address from the

- Queue
- Stack
- Program counter
- Pointer

III. The external system bus architecture is created using from _____ architecture:

- Pascal
- Dennis Ritchie
- Charles Babbage
- Von Neumann

IV. The processor 80386/80486 and the Pentium processor uses _____ bits address bus:

- 16
- 64
- 32
- 36

V. CS Stand for.

- Chip select
- Control select
- Cable select
- Computer select

VI. How many address ways of transferring data between the microprocessor and a physical I/O device?

- 1
- 2
- 3
- 4

VII. The CPU completes yields control of the bus to the DMA controller via:

- DMA acknowledge signal
- DMA integrated signal
- DMA implicitly signal
- None of these

VIII. Which are the parts of architecture of 8086?

- BIU
- 16 bit Accumulator
- EU
- All the above

IX. Which causes the microprocessor to immediately terminate its present activity?

- RESET signal
- INTERUPT signal
- RESET and INTERUPT signal
- None of these

X. INTR: it implies the_____ signal

- INTERUPT Request
- INTERUPT Reset
- INTERUPT Right
- INTERUPT wrong

XI. What is the control word in BSR mode to set PC4?

- 1xxx1000
- 0xxx1000
- 1xxx1001
- 0xxx1001

XII. Intel 8253/54 is a.

- USART
- DMA.
- TIMER
- Relative

XIII. If the operating frequency is 5MHz then time required for 1 T-state is:

- 0.1 μsec
- 0.2 μsec
- 0.3 μsec
- 0.4 μsec.

XIV. What is the addressing mode used in instruction MOV M,C?

- Direct
- Indirect
- Immediate
- Indexed

XV. CMP R/M instruction of 8085 microprocessor set zero flag

- A > R/M
- A=R/M
- A < R/M
- None of these

XVI. 8254 can handle input upto

- 5MHz
- 10MHz
- 15MHz
- 20MHz

SECTION-B (Attempt any four) **(4*3=12)**

I. Write a program to add the contents of the memory location 2000H: 0500H with contents of 3000H: 0600H memory location and store the result in 5000H: 0700H?

II. What is need of addressing modes in microprocessor? Give overview of microprocessor family.

III. Explain the following instructions with suitable example. Also indicate the flag condition.
i) ADI ii) DAA iii) RAR

IV. List the internal registers of 8086 microprocessor with their abbreviations.

V. Discuss relationship between Instruction cycle, Machine cycle and T-state with diagram.

SECTION-C (Attempt any four) **(4*5=20)**

I. Draw and discuss the block diagram of the internal architecture of PIC 8259.

II. What is the necessity of the programmable interval timer? Discuss the organization of Timer with a functional block diagram.

III. What is the use of 8237 IC? Discuss mode register and various modes of 8237.

IV. Draw the internal architecture of 8255 and mention the control word format of BSR Mode.

V. Write short notes on it,
i) Programmed I/O ii) Memory Mapped I/O

Uni. Roll No:-.....

SET-D

GLA University, Mathura

Course: -B.Tech. II-Year, End-Term (Even Sem.) Examination, 2012-13

Subject: - OPERATING SYSTEM (CSE-221)

Time:-2.30Hour M.M:-40

Notes:-

1. Answer all questions from Group A (which is) compulsory, Any Four from Group B and Any Four from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group-A (All questions are compulsory) **16X0.5=08**

I. System calls are usually invoked by using

- An indirect jump
- A Software interrupt
- Polling
- A privileged instruction

II. If the length of time slice is made more than the burst time of the process with highest burst time then round robin scheduling behaves similar to:

- Priority scheduling
- FCFS scheduling
- SJF scheduling
- None of these

III. Locality of reference implies that the page reference being made by a process

- Will always be to the page used in the previous page reference
- Is likey to be to one of the pages used in the last few page references
- Will always be to one of the pages existing in memory
- Will always lead to page fault

IV. Dirty bit of a page in a page table

- Allows only read on a page
- Helps avoid unnecessary writes on a paging device
- Helps maintian LRU information
- Used for protection

V. A counting semaphore was initialized to 10. Then 6P (wait) operations and 4V (signal) operations were performed on semaphore. What is the resulting value of semaphore?

- 2
- 12
- 8
- 0

VI. A computer has 6 tape drives, with 'n' processes competing for them. Each process may need two drives. What is the maximum value of 'n' for the system to be deadlock free?

- 4
- 6
- 5
- 3

VII. Which of the following scheduling algorithm is non-preemptive?

- Round-Robin
- FCFS
- Multilevel queue scheduling
- Multilevel queue with feedback

VIII. Which of the following need not necessarily be saved on a context saved between processes?

- General purpose registers
- Translation Look-Aside Buffer
- Program Counter
- All of the above

IX. Suppose an OS uses variable length partitions for memory management. At some particular time a running process occupies a partition between physical address 1000 and 20000. What is the physical address for virtual address 3000?

- 23000
- 25000
- 13000
- 33000

X. Which of the following file allocation technique suffers from external fragmentation?

- Contiguous allocation
- Indexed allocation
- Linked allocation
- Both Contiguous and Indexed allocation

XI. A virtual memory system has 32-bit virtual and physical address space with page size of 2 KB. How many bits will be needed for index and off-set of page?

- 21, 11
- 32, 32
- 32, 8
- 11, 21

XII. What information does the presence of a cycle in resource allocation graph give?

- Deadlock will always occur
- Deadlock will never occur
- Deadlock may or may not occur
- All of the above

XIII. The correct matching for the following pairs:

(A) Disk Scheduling
(B) Batch Processing
(C) Time Sharing
(D) Interrupt processing

(1) Round Robin
(2) SCAN
(3) LIFO
(4) FIFO

- A-4, B3, C-2 and D-1
- A-2, B-4, C-1 and D-3
- A-3, B-4, C-2, and D-1
- A-2, B-1, C-4 and D-3

XIV. Listed below are some OS abstractions (in left column) and the hardware components (in right column). Match the correct pairs:

(P) Thread
(Q) Virtual Address Space
(R) File System
(S) Signal

(1) Interrupt
(2) Memory
(3) CPU
(4) Disk

- P-3, Q-2, R-4, S-1
- P-1, Q-2, R-3, S-4
- P-2, Q-4, R-3, S-1
- P-4, Q-1, R-2, S-3

XV. Using large block size in a fixed block size file system leads to:

- Poorer disk throughput and poorer disk space utilization
- Better disk throughput and better disk space utilization
- Poorer disk throughput but better disk space utilization
- Better disk throughput but poor disk space utilization

XVI. In which of the following page replacement policies does Belady's anomaly occur?

- LFU
- NRU
- FIFO
- LRU

GLA University, Mathura **SET A**

End Term Examination, 2012-13

Course: - B.Tech. II-Year, IV-Sem.

Subject: - Computer Graphics (CSE-223) Univ. Roll No.....

Time:-2 Hrs. 30 Minutes Max. Marks 40

Notes:- 1) Answer all questions , Section A is compulsory, Attempt any **four** from Section B and any **four** from section C.

2) All parts of a question (a, b, etc.) should be answered at one place.

3) Answer should be brief and to-the-point and be supplemented with neat sketches.

4) Any missing or wrong data may be assumed suitably giving proper justification.

5) Figures on the right-hand side margin indicate full marks.

Section A (16 x0.5=08)

Q.I Seed point is used in

• Boundary filling •Flood fill • Scan line filling •Inside-Outside test

Q.II .....parametric continuity calculate the tangent at two curve section meeting point.

• Zero order •First order •Second order • Higher than two order

Q.III Draw back of DDA line drawing algorithm is, it uses

•Real numbers •Integer numbers •More memory space •None

Q.IV Lossy Compression Techniques reduce the size of .....data.

•Audio •Video •Audio and Video • Text

Q.V GKS is :

• Reliable • Portable • Non-Portable • None of these

Q.VI An LZW string table consists of entries from

•0-4096 • 256-4096 • 0-256 • 0 to ∞

Q.VII The resolution of a printer is measured in terms of:

• pixel density • dot pitch • herts • dpi

Q.VIII DVI is a

• Graphics standard • Graphics software • Multimedia Standard • None

Q.IX A process of integrated multimedia development is known as:

• Authoring • Morphing • Tools • None of these

Q.X In parallel projection, the center of projection is at

• infinity • finite distance • x-axis •y-axis

Q.XI MPEG stands for:

• Mass Picture Expert Group •Motion Picture Expert Group

• More Picture Expert Group • Mini Picture Expert Group

Q.XII The first multimedia desktop video standard for IBM-compatible personal computers

• GKS • DVI • PHIGS • All of them

Q.XIII Z- buffer algorithm is used for

•Transparent surfaces • Opaque surfaces • intersection surfaces • None

Q.XIV The distance between the pixels on a screen is called:

• OCR • LCD • Refresh rate • Dot pitch

Q.XV In what illumination model, the light reflected from surface is equal in all the directions or viewing angle?

• Ambient Light • Specular Reflection

• Diffuse Reflection • Phong Reflection

Q.XVI MIDI stands for:

•Many Instruments Digital Inputs • Musical instruction Digital Interface

• Music Interface Digital Instructions •Musical Instrumental Digital Interface

Section B (4 x 3=12)

Q. I Define window and viewport. What do you understand by viewing transformation? [1+2]

Q.II Explain the working of color CRT with shadow mask method. [3]

Q. III Show that rotation about x axis followed by a rotation about y axis is not commutative.[3]

Q. IV Rotate a triangle (0,0), (1,0), (1,1) with respect to the origin anticlockwise by an angle 30° and then translate it 3 units to the left and 5 unit to down. [3]

Q.V Write Bresenham's line drawing algorithm. Consider a line from (0,0) to (6, 6) by using Bresenham's line drawing algorithm rasterizing this line. [1.5+1.5]

Section C (4 x 5=20)

Q. I What is the importance of hidden line and surface removal algorithm? Discuss the mechanism of Z-buffer surface removal algorithm and differentiate it with A-buffer surface removal algorithm. [1+3+1]

Q.II Write short note on JPEG? Explain the steps for JPEG encoding. [2.5+2.5]

Q.III What do you mean by Graphics Standards? Explain GKS with its primitives and also explain the role of metafiles in GKS. [1+4]

Q.IV Explain multimedia with its components and also explain five applications of Multimedia. [0.5+2+2.5]

Q.V Suppose a message composed of six symbol need to be communicated over a communication channel. The symbols and their probabilities of occurrence are shown in the given table. Generate the Shanon- Fano code for each symbol. Compute the average number of bits required to communicate a symbol and estimate the amount of redundancy per symbol. Evaluate the performance over traditional ASCII codes. [5]

Symbols	**Probability**
A	0.15
B	0.25
C	0.30
D	0.15
E	0.05
F	0.10

SET-A Uni. Roll No:-.....

GLA University, Mathura

Course: -B.Tech. II-Year **End-Term (Even Sem.) Examination, 2012-13** **M.M: 40**

Time:-2Hrs 30 Minutes **Subject: - Computer Networks (CSE-224)**

Notes:-

1. Answer all questions from Group A (which is) **compulsory**, **Any Four** from Group B and **Any Four** from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification.
5. Figures on the right-hand side margin indicate full marks.

Group-A(All questions are compulsory) [16x0.5=08]

Ques.1 In OSI network architecture, the dialogue control and token management are responsibility of

- data link layer
- network layer
- session layer
- transport layer

Ques.2 The interactive transmission of data within a time sharing system may be best suited to

- simplex lines
- half-duplex lines
- full duplex lines
- biflex-lines

Ques.3 Which of the following statement is incorrect?

- The difference between synchronous and asynchronous transmission is the clocking derived from the data in synchronous transmission.
- Half duplex line is a communication line in which data can move in two directions, but not at the same time.
- Teleprocessing combines telecommunications and DP techniques in online activities
- Batch processing is the prefered processing mode for telecommunication operation.

Ques.4 The transmission signal coding method of TI carrier is called

- Bipolar
- NRZ
- Manchester
- Binary

Ques.5 Which of the address is multicast

- Class E
- Class B
- Class C
- Class D

Ques.6 When a destination host transport layer receives data from the network layer, it unambiguously identifies the appropriate process to pass the data to by using a triplet consisting of:

- source port #, destination IP address, and source IP address
- destination port #, source port #, process ID#
- destination port #, source port #, destination IP address
- destination port #, source port #, source IP address

Ques.7 Mail services are available to network users through the _____ layer.

- Data link
- Physical
- Transport
- Application

Ques.8 As the data packet moves from the upper to the lower layers, headers are _____.

- Added
- Removed
- Rearranged
- Modified

Ques.9 The process of a web server adding a TCP header to a web page, followed by adding an IP header, and then a data link header and trailer is an example of what?

- Data encapsulation
- Same-layer interaction
- The OSI model
- All the options given are correct

Ques.10 Which of the following protocols are examples of TCP/IP network access layer protocols?

- HTTP
- IP
- SMTP
- PPP

Ques.11 The process of HTTP asking TCP to send some data and make sure that it is received correctly is an example of what?

- Same-layer interaction
- Adjacent-layer interaction
- The OSI model
- All the options are correct

Ques.12 What is the Hamming distance between 001111 and 010011?

- 1
- 2
- 3
- 4

Ques.13 Which of the following terms is used specifically to identify the entity that is created when encapsulating data inside data link layer headers and trailers?

- Chunk
- Segment
- Frame
- Packet

Ques.14 Number of links to connect n nodes in a mesh topology is_____

- n
- n(n-1)
- n(n-1)/2
- n(n-2)/2

Ques.15 Which of the following protocols are examples of TCP/IP transport layer protocols?

- HTTP
- IP
- UDP
- FTP

Ques.16 Which OSI layer defines the standards for data formats and encryption?

- Layer 4
- Layer 5
- Layer 6
- Layer 7

Group-B(Answer any four) [4x3=12]

Ques.1Define various ways to map physical to logical address.

Ques.2Explain 1-persistent, p-persistent and 0-persistent CSMA giving strong and weak points of each.

Ques.3Compare IPv6 and IPv4 characteristics.

Ques.4Define any one Traffic Shaping mechanisms to control the rate of the traffic sent to the network. In a leaky bucket used to control liquid flow, how many gallons of liquid are left in the bucket if the output rate is 6 gal/min, there is an input burst of 100 gal/min for 11s, and there is no input for 49 s?

Ques.5 Explain Distance Vector routing protocol and count to infinity problem with its solution.

Group-C (Answer any four) [4x5=20]

Ques.1 Explain MIME.How does MIME enhance SMTP?

Ques.2 Write sort notes on

a) DNS b)WWW

Ques.3 Describe the TCP Header Format.

Ques.4 What do you mean by cryptography? Explain the concept of symmetric and asymmetric key management with suitable example.

Ques.5 Explain TCP connection management mechanism.

GLA UNIVERSITY MATHURA

Uni. Roll No:-.....

SET-C

GLA University, Mathura

Course: -B.Tech. II-Year, End-Term (Even Sem.) Examination, 2012-13

Subject: - Environmental Studies (CEE-201)

Time:-2.30 Hour **M.M:-40**

Notes:-

1. Answer all questions from Group A (which is) **compulsory**, **Any Four** from Group B and **Any Four** from Group C.
2. All questions of the particular group should be answered collectively at one place. All parts of a question (a, b, etc.) should be answered at one place.
3. Answer should be brief and to-the-point and be supplemented with neat sketches.
4. Any missing or wrong data may be assumed suitably giving proper justification. Figures on the right-hand side margin indicate full marks.

Group-A (All questions are compulsory) 16 X 0.5=08

It contains 16 questions. Choose and write the correct answer.

I. Who is known as water Man of India -

- Sh Sundrlal Bahuguna
- Smt. Medha Ptekar
- Rajendra Singh
- Ms Arundhati Ray

II. The Air (prevention and control of pollution) Act was passed in

- 1981
- 1980
- 1972
- 1974

III. Normal average thickness of atmospheric ozone layer across the globe around-

- 200 DU
- 500 DU
- 400 DU
- 300 DU

IV. Population pyramids are useful to -

- Express the pollution growth
- Indicate the birth rate
- Express the age-sex distribution of a population
- Indicate the death rate

V. Which of the following is relevant to environmental ethics -

- Value of a humanly restored environment compared with originally natural environment
- Rights species other than human beings on this planet.
- Obligation of companies in environmental protection
- All of above

VI. Which Biomedical waste may be disposed off by-

- Incineration
- Autoclaving and land filling
- Both as above
- None of the above

VII. Which of the following is a secondary air pollutant-

- O_3
- O_2
- CO
- CO_2

VIII. What is the maximum allowable concentration of fluoride in potable water in milligrams per liter?-

- 1.0
- 1.5
- 1.25
- 1.75

IX. Air pollution from automobiles can be controlled by-

- Electrostatic precipitator
- Wet collector (scrubber)
- Catalytic converter
- All of the above

X. Who introduced the concept of biodiversity hotspot?

- Christopher Columbus
- Norman Myers
- WWF
- Charles Darwin

XI. Lion-Tailed Macaque is found in-

- Western Ghats
- Eastern Ghats
- Caucasus
- Western Himalaya

XII. In an aquatic ecosystem phytoplankton can be considered as a

- Producer
- Consumer
- Saprotrophic organisms
- Macroconsumer

XIII. The largest reservoir of nitrogen in our planet-

- Oceans
- Atmosphere
- Biosphere
- Fossil fuel

XIV. India has the world's largest share of which of the following?

- Manganese
- Mica
- Copper
- Diamond

XV. Which of the following is not an ideal solution for tackling the water crisis?

- Drilling large number deep bore wells
- Population growth control
- Water conservation in irrigation
- Water pollution control

XVI. Which of the following conceptual spheres of the environment is having the least storage capacity for matter?

- Atmosphere
- Lithosphere
- Hydrosphere
- Biosphere

Group-B (Answer any Four) 4 x 3=12

I What are the major scopes of environmental science?

II Describe briefly the effect of mining on environment.

III Write salient features of balanced ecosystem.

IV Write the various applications of solar energy.

V Mention the effects of noise pollution on human being.

Group-C (Answer any Four) 4 x 5=20

I Define environmental education and write its objectives and guidelines for excellence in environmental education.

II Why EIA is needed to establish a new project? Describe the steps involved in EIA.

III. Discuss the role of government in environmental protection.

IV Write major environmental laws and discuss the salient feature of Water (prevention and control of pollution) Act, 1974.

V Write short notes on any two of the following.

(a) Ecological foot print **(b)** Functions of NGO **(c)** Environmental awareness